

GPT-5.5 Merged Evaluation Review Samples

Random test-set samples from Physics and FrontierPhysics. Each sample shows the problem statement, extracted boxed answer, reference final solution, and GPT-5.5 judge rationale.

FRONTIERPHYSICS · TEST · MERGED MODE

Score 1.000

1. textbook/11.9.2

PROBLEM STATEMENT

Ornstein 6Uhlenbeck process

The Ornstein 6Uhlenbeck process is the harmonic oscillator of nonequilibrium dynamics. It describes stochastic evolution in cases where the averaged dynamics of a Langevin particle is described by a linear first-order differential equation. Realizations include the dynamics of a particle subject only to friction (but no external forces), or that of an overdamped particle in a harmonic external potential. The dissipative dynamics close to extremal potential points is often described by variants of this process.

(a) Consider a Brownian particle in one dimension whose velocity is governed by the stochastic differential equation

$$\dot{v} + \gamma v = \frac{f(t)}{m},$$

(11.83)

where f is a Langevin force, and $\langle f(t)f(t') \rangle = A\delta(t - t')$. Determine the general solution of Eq.

(11.83) for the velocity with the initial condition $v(0) = v_0$, and calculate $\langle v(t) \rangle$ and $\langle v^2(t) \rangle$. Using the long-time limit of the latter, identify the coefficient A with the help of the equipartition theorem.

(b) Consider the time-dependent velocity distribution $p(v, t)$. It is governed by the Fokker-Planck equation

$$\left[\frac{\partial}{\partial t} + \frac{\partial}{\partial v} a_1(v) - \frac{1}{2} \frac{\partial^2}{\partial v^2} a_2(v) \right] p(v, t) = 0.$$

(11.84)

With the help of Eq. (11.84), give equations for $\partial_t \langle v \rangle$ and $\partial_t \langle v^2 \rangle$ and identify the coefficients a_1 and a_2 , e.g., by using the results of part (a). Give the corresponding Fokker-Planck equation for the generating function $g(k, t) = \int dv e^{-ikv} p(v, t)$.

(c) The first-order partial differential equation (11.85) can be solved, for example with the method of characteristics. A general solution is of the form

$$g(k, t) = e^{-\frac{\gamma k^2}{2m}} \phi(ke^{-\gamma t})$$

Determine the function ϕ from the initial condition $p(v, t = 0) = \delta(v - v_0)$ and determine the distribution $p(v, t)$. Give the short- and long-time limits of your result.

Equation (11.83): $\dot{v} + \gamma v = \frac{f(t)}{m}$.

Equation (11.84):

$$\partial_t p + \partial_v (a_1(v)p) - \frac{1}{2} \partial_v^2 (a_2(v)p) = 0.$$

Equation (11.85):

$$\left(\partial_t + \gamma k \partial_k + \frac{\gamma T}{m} k^2 \right) g(k, t) = 0.$$

EXTRACTED ANSWER

$$\begin{aligned} \text{(a)} \quad v(t) &= v_0 e^{-\gamma t} + \frac{1}{m} \int_0^t ds e^{-\gamma(t-s)} f(s), \\ \langle v(t) \rangle &= v_0 e^{-\gamma t}, \\ \langle v^2(t) \rangle &= v_0^2 e^{-2\gamma t} + \frac{A}{2\gamma m^2} (1 - e^{-2\gamma t}). \text{ Since } \lim_{t \rightarrow \infty} \langle v^2 \rangle = \frac{T}{m}, \quad A = 2\gamma m T. \\ \text{(b)} \quad \partial_t \langle v \rangle &= \langle a_1(v) \rangle, \quad \partial_t \langle v^2 \rangle = 2 \langle v a_1(v) \rangle + \langle a_2(v) \rangle. \\ a_1(v) &= -\gamma v, \quad a_2(v) = \frac{A}{m^2} = \frac{2\gamma T}{m}. \\ \partial_t p &= \gamma \partial_v (vp) + \frac{\gamma T}{m} \partial_v^2 p, \\ \left(\partial_t + \gamma k \partial_k + \frac{\gamma T}{m} k^2 \right) g(k, t) &= 0, \quad g(k, t) = \int dv e^{-ikv} p(v, t). \\ \text{(c)} \quad g(k, t) &= e^{-Tk^2/(2m)} \phi(k e^{-\gamma t}), \quad g(k, 0) = e^{-ikv_0}, \\ \phi(q) &= \exp \left[-iqv_0 + \frac{Tq^2}{2m} \right], \\ g(k, t) &= \exp \left[-ikv_0 e^{-\gamma t} - \frac{T}{2m} (1 - e^{-2\gamma t}) k^2 \right], \\ p(v, t) &= \sqrt{\frac{m}{2\pi T(1 - e^{-2\gamma t})}} \exp \left[-\frac{m(v - v_0 e^{-\gamma t})^2}{2T(1 - e^{-2\gamma t})} \right]. \\ t \rightarrow 0: \quad p(v, t) &\rightarrow \delta(v - v_0), \text{ more precisely a narrow Gaussian with mean } v_0 e^{-\gamma t} \text{ and variance } \frac{T}{m}(1 - e^{-2\gamma t}) \simeq \frac{2\gamma T}{m} t. \\ t \rightarrow \infty: \quad p(v, t) &\rightarrow \sqrt{\frac{m}{2\pi T}} \exp \left(-\frac{mv^2}{2T} \right). \end{aligned}$$

FINAL SOLUTION

(a) Integrating the equation, one obtains

$$v(t) = e^{-\gamma t} \left[v_0 + \int_0^t dt' \frac{f(t')}{m} e^{\gamma t'} \right].$$

Averaging over the Langevin force with $\langle f \rangle = 0$ gives $\langle v(t) \rangle = v_0 e^{-\gamma t}$. For the second moment,

$$\langle v^2(t) \rangle = e^{-2\gamma t} \left[v_0^2 + \int_0^t dt' dt'' \frac{\langle f(t') f(t'') \rangle}{m^2} e^{\gamma(t'+t'')} \right] = e^{-2\gamma t} \left[v_0^2 + \frac{A}{2\gamma m^2} (e^{2\gamma t} - 1) \right].$$

In the long-time limit, $\langle v^2(t) \rangle \rightarrow \frac{A}{2\gamma m^2}$. Equipartition, $\frac{1}{2} m \langle v^2 \rangle = \frac{1}{2} T$, then fixes $A = 2\gamma m T$.

(b) Using the Fokker-Planck equation and integrating by parts,

$$\begin{aligned}\partial_t \langle v \rangle &= \langle a_1(v) \rangle, \\ \partial_t \langle v^2 \rangle &= \langle 2va_1(v) \rangle + \langle a_2(v) \rangle.\end{aligned}$$

Comparison with part (a) implies $a_1(v) = -\gamma v$ and $\langle a_2 \rangle = 2\gamma T/m$, which for this linear process means $a_2(v) = 2\gamma T/m$. Fourier transforming $p \mapsto g(k, t)$ yields

$$\left[\partial_t + \gamma k \partial_k + \frac{\gamma T}{m} k^2 \right] g(k, t) = 0,$$

which is Eq. (11.85).

(c) Method of characteristics for (11.85): characteristics satisfy $dk/dt = \gamma k$ so $k(t) = k_0 e^{\gamma t}$. Along them,

$$\frac{dg}{dt} = -\frac{\gamma T}{m} k^2 g \Rightarrow g(k, t) = \exp\left(-\frac{T}{2m} k^2\right) \phi(k e^{-\gamma t}).$$

Initial condition $p(v, 0) = \delta(v - v_0)$ implies $g(k, 0) = e^{-ikv_0}$, hence $\phi(k) = e^{-ikv_0}$. Therefore

$$g(k, t) = \exp\left[-ikv_0 e^{-\gamma t} - \frac{T}{2m} k^2 (1 - e^{-2\gamma t})\right].$$

Inverse Fourier transform gives the Ornstein-Uhlenbeck velocity density

$$p(v, t) = \left(\frac{2\pi T}{m}(1 - e^{-2\gamma t})\right)^{-1/2} \exp\left[-\frac{(v - v_0 e^{-\gamma t})^2}{2(T/m)(1 - e^{-2\gamma t})}\right].$$

Short-time limit $t \ll \gamma^{-1}$: a Gaussian centered at v_0 with variance $2(T/m)\gamma t$; long-time limit $t \gg \gamma^{-1}$: stationary Maxwell distribution $p(v) = \sqrt{m/(2\pi T)} \exp(-mv^2/(2T))$.

JUDGE RATIONALE

- **(a)** Gives the correct integrated velocity solution, mean, second moment, long-time limit, and uses equipartition to identify $A = 2\gamma m T$.
- **(b)** Correctly states the moment equations, identifies $a_1(v) = -\gamma v$ and $a_2(v) = 2\gamma T/m$, and gives the equivalent Fokker-Planck and generating-function equations.
- **(c)** Correctly applies the delta initial condition to obtain ϕ and $g(k, t)$, gives the correct Gaussian Ornstein-Uhlenbeck distribution, and states the correct short- and long-time limits.

2. textbook/12.3

PROBLEM STATEMENT

Consider a nonideal gas obeying a modified van der Waals equation of state

$$\left(P + \frac{a}{v^n}\right)(v - b) = RT \quad (n > 1).$$

Examine how, of this system, depend on the number n :

- (a) the critical constants P_c , v_c , and T_c .
- (b) the critical exponents β , γ , γ' , and δ .

EXTRACTED ANSWER

$$\begin{aligned} \text{(a)} \quad v_c &= \frac{n+1}{n-1}b, \quad T_c = \frac{4na}{Rb^{n-1}} \frac{(n-1)^{n-1}}{(n+1)^{n+1}}, \quad P_c = \frac{a}{b^n} \frac{(n-1)^{n+1}}{(n+1)^{n+1}}. \\ \text{(b)} \quad \beta &= \frac{1}{2}, \quad \gamma = 1, \quad \gamma' = 1, \quad \delta = 3. \end{aligned}$$

FINAL SOLUTION

The given equation of state (for one mole) of the gas is

$$P = \frac{RT}{v-b} - \frac{a}{v^n} \quad (n > 1). \quad (1)$$

Equating $(\partial P/\partial v)_T$ and $(\partial^2 P/\partial v^2)_T$ to zero, we get

$$v_c = \frac{n+1}{n-1}b, \quad T_c = \frac{4n(n-1)^{n-1}}{(n+1)^{n+1}} \frac{a}{b^{n-1}R}.$$

Equation (1) then gives

$$P_c = \left(\frac{n-1}{n+1}\right)^{n+1} \frac{a}{b^n},$$

whence $\frac{RT_c}{P_c} v_c = \frac{4n}{n^2-1}$.

To determine the critical behaviour, set $P = P_c(1 + \pi)$, $v = v_c(1 + \psi)$, $T = T_c(1 + t)$. A standard expansion yields

$$\pi \approx \frac{2n}{n-1}t - \frac{n(n+1)^2}{12}\psi^3 + \frac{n(n+1)}{n-1}t\psi,$$

so that along $t = 0$, $\pi \sim -\psi^3$; for $t < 0$, $\psi_{1,3} \sim \pm|t|^{1/2}$; and $-(\partial\psi/\partial\pi)_t \sim t^{-1}$ above T_c and $\sim |t|^{-1}$ below along the stable branches. Thus the critical exponents are the same as for the van der Waals gas for all $n > 1$:

$$\beta = \frac{1}{2}, \quad \gamma = 1, \quad \gamma' = 1, \quad \delta = 3,$$

while the critical constants depend on n as given above.

JUDGE RATIONALE

- **(a)** Gives the correct n -dependent critical constants v_c , T_c , and P_c matching the reference solution.
- **(b)** Gives the correct critical exponents $\beta=1/2$, $\gamma=1$, $\gamma'=1$, and $\delta=3$ for all $n>1$.

3. textbook/5.5

PROBLEM STATEMENT

In the first approximation the partition function of a system of N noninteracting, indistinguishable particles is given by

$$Q_N(V, T) = \frac{1}{N! \lambda^{3N}} Z_N(V, T),$$

where

$$Z_N(V, T) = \int \exp \left\{ -\beta \sum_{i < j} v_s(r_{ij}) \right\} d^{3N} r,$$

$v_s(r)$ being the statistical potential ([5.5.28]). Evaluate the first-order correction to the equation of state of this system.

EXTRACTED ANSWER

$$\frac{PV}{NkT} = 1 \mp \frac{N\lambda^3}{2^{5/2}V} \quad \text{with } - \text{ for bosons and } + \text{ for fermions, i.e. } \frac{PV}{NkT} = 1 - \frac{n\lambda^3}{2^{5/2}} \text{ for bosons, } 1 + \frac{n\lambda^3}{2^{5/2}} \text{ for fermions, } n = N/V.$$

FINAL SOLUTION

By eqn. (5.5.17), we have

$$Q_N(V, T) \equiv \text{Tr} \left(e^{-\beta \hat{H}} \right) = \frac{1}{N! \lambda^{3N}} Z_N(V, T),$$

where

$$Z_N(V, T) = \int \sum_P \{ \dots \} d^{3N} r. \quad (1)$$

In the zeroth approximation, $\sum_P = 1$ (see eqn. (5.5.19)), so $Z_N(V, T) = V^N$. In the first approximation, if λ is much smaller than the mean interparticle distance, we may write

$$\sum_P \approx \prod_{i < j} \left(1 \pm e^{-2\pi r_{ij}^2 / \lambda^2} \right) = \prod_{i < j} e^{-\beta v_s(r_{ij})} = \exp \left\{ -\beta \sum_{i < j} v_s(r_{ij}) \right\}, \quad (2)$$

which leads directly to the desired result for Q_N in terms of Z_N .

For the second part, substitute (2) into (1) and integrate over the position coordinates. On assembling contributions from all pairs of particles, one finds

$$Z_N(V, T) = V^N \pm \frac{N(N-1)}{2} V^{N-2} \frac{\lambda^3}{2^{3/2}}.$$

For $N \gg 1$ and $N\lambda^3 \ll V$,

$$Z_N(V, T) = V^N \left[1 \pm \frac{N^2 \lambda^3}{2^{5/2} V} \right] \approx V^N \left[1 \pm \frac{N \lambda^3}{2^{5/2} V} \right]^N.$$

It follows that

$$\ln Q_N(V, T) \approx -N \ln N + N + N \ln \left(\frac{V}{\lambda^3} \right) + N \left(\pm \frac{N \lambda^3}{2^{5/2} V} \right),$$

whence

$$\frac{P}{kT} \equiv \left(\frac{\partial \ln Q_N}{\partial V} \right)_{N,T} \approx \frac{N}{V} \mp \frac{N^2 \lambda^3}{2^{5/2} V^2} = \frac{1}{v} \mp \frac{1}{2^{5/2}} \frac{\lambda^3}{v^2},$$

where $v = V/N$. This gives the first-order quantum-statistical correction to the ideal-gas equation of state (upper sign for bosons, lower for fermions).

JUDGE RATIONALE

- **(a)** Correct first-order equation of state: $PV/(NkT) = 1 \mp N\lambda^3/(2^{5/2}V)$, equivalently $1 \mp n\lambda^3/2^{5/2}$, with negative correction for bosons and positive for fermions.

4. textbook/5.6

PROBLEM STATEMENT

The Manning transition: A linear polyelectrolyte (rod) of length L and radius a carries N positive charges left on the rod, while N negatively charged counter-ions are in solution (water). Ignoring Coulomb repulsion among counter-ions, each counter-ion moves in the potential $\phi(r) = -2(Ne/L) \ln(r/L)$ (cylindrical geometry), with Hamiltonian

$$\mathcal{H} = \sum_{i=1}^N \left[\frac{p_i^2}{2m} + 2e^2 n \ln(r/L) \right], \quad n \equiv N/L.$$

The container is a cylinder of radius R and height L .

- Compute the canonical partition function Z in terms of T, n, R, a .
- Compute the single-particle radial distribution $p(r)$ and its first moment $\langle r \rangle$.
- Identify the transition temperature in the limit $R \gg a$. Indicate the scaling of $\langle r \rangle$ with R or a .
- Compute the pressure on the cylindrical wall $r = R$ for $R \gg a$, all T .
- Now include ion-ion repulsions approximately by allowing N_1 ions to condense on the rod and $N_2 = N - N_1$ to remain free with Hamiltonian $\sum_{i=1}^{N_2} [p_i^2/(2m) + 2e^2 n_2 \ln(r/L)]$, where $n_2 = N_2/L$. Guess the equilibrium N_2^* .

EXTRACTED ANSWER

$$\begin{aligned} \lambda_T &= \frac{h}{\sqrt{2\pi m k_B T}}, \quad q = \frac{e^2 n}{k_B T}, \quad N = nL. \\ \text{(a) } Z &= \frac{1}{N!} \left[\frac{2\pi L^3}{\lambda_T^3} A_q \right]^N, \quad A_q = \begin{cases} \frac{(R/L)^{2(1-q)} - (a/L)^{2(1-q)}}{2(1-q)}, & q \neq 1, \\ \ln(R/a), & q = 1. \end{cases} \\ \text{(b) } p(r) &= \frac{r^{1-2q}}{\int_a^R r'^{1-2q} dr'}, \quad a \leq r \leq R, \\ p(r) &= \begin{cases} \frac{2(1-q)r^{1-2q}}{R^{2-2q} - a^{2-2q}}, & q \neq 1, \\ \frac{1}{r \ln(R/a)}, & q = 1, \end{cases} \\ \langle r \rangle &= \begin{cases} \frac{2(1-q)}{3-2q} \frac{R^{3-2q} - a^{3-2q}}{R^{2-2q} - a^{2-2q}}, & q \neq 1, \\ \frac{R-a}{\ln(R/a)}, & q = 1, \\ \frac{aR}{R-a} \ln(R/a), & q = 3/2. \end{cases} \\ \text{(c) } T_c &= \frac{e^2 n}{k_B} (q = 1). \text{ For } R \gg a: \quad \langle r \rangle \sim \begin{cases} R, & T > T_c, \\ R/\ln(R/a), & T = T_c, \\ a(R/a)^{3-2q}, & 2T_c/3 < T < T_c, \\ a \ln(R/a), & T = 2T_c/3, \\ a, & T < 2T_c/3. \end{cases} \\ \text{(d) } P &= \frac{k_B T}{2\pi R L} \frac{\partial \ln Z}{\partial R} \sim \begin{cases} \frac{nk_B T(1-q)}{\pi R^2}, & q < 1, \\ \frac{nk_B T}{2\pi R^2 \ln(R/a)}, & q = 1, \\ \frac{nk_B T(q-1)}{\pi R^2} \left(\frac{a}{R}\right)^{2q-2}, & q > 1. \end{cases} \\ \text{(e) } N_2^* &= \min\left(N, \frac{Lk_B T}{e^2}\right), \quad N_1^* = N - N_2^*, \text{ i.e. the free ions self-adjust to } e^2(N_2^*/L)/(k_B T) = 1 \text{ when condensed.} \end{aligned}$$

FINAL SOLUTION

(a) Each particle contributes a momentum factor λ^{-3} with $\lambda = h/\sqrt{2\pi m k_B T}$. In cylindrical coordinates, the spatial integral per particle gives $\int_a^R (2\pi L) r dr (r/L)^{-2e^2 n/(k_B T)}$. Hence

$$Z = \frac{1}{N!} \left(\frac{1}{\lambda^3} \int_a^R 2\pi L r dr (r/L)^{-2e^2 n/(k_B T)} \right)^N = \left(\frac{2\pi e}{n \lambda^3} \right)^N L^{2Ne^2 n/(k_B T)} \left[\frac{R^{2(1-e^2 n/(k_B T))} - a^{2(1-e^2 n/(k_B T))}}{2(1-e^2 n/(k_B T))} \right]^N.$$

(up to overall constants not affecting thermodynamics in (c,d)).

(b) The one-body distribution is

$$p(r) = \frac{r e^{-[2e^2 n/(k_B T)] \ln(r/L)}}{\int_a^R dr r e^{-[2e^2 n/(k_B T)] \ln(r/L)}} = 2 \left(1 - \frac{e^2 n}{k_B T} \right) \frac{r^{1-2e^2 n/(k_B T)}}{R^{2(1-e^2 n/(k_B T))} - a^{2(1-e^2 n/(k_B T))}},$$

normalized on $[a, R]$. The mean radius is

$$\langle r \rangle = \int_a^R dr r p(r) = \frac{2k_B T - 2e^2 n}{3k_B T - 2e^2 n} \frac{R^{3-2e^2 n/(k_B T)} - a^{3-2e^2 n/(k_B T)}}{R^{2-2e^2 n/(k_B T)} - a^{2-2e^2 n/(k_B T)}}.$$

(c) For $R \gg a$: If $e^2 n/(k_B T) > 1$ (low T), $p(r) \propto r^{1-2e^2 n/(k_B T)}$ is normalizable as $R \rightarrow \infty$ and $\langle r \rangle \propto a$ (condensed on the rod). If $e^2 n/(k_B T) < 1$ (high T), $\langle r \rangle \propto R$ (counter-ions spread in the container). The transition is at

$$T_c = \frac{e^2 n}{k_B}.$$

(d) The wall pressure follows from $P = (k_B T)/(2\pi RL) \partial \ln Z / \partial R$. For $T < T_c$, Z becomes R -independent in the $R \gg a$ limit, so $P = 0$. For $T > T_c$, one finds

$$P = \frac{k_B T}{2\pi RL} \cdot \frac{2N(1 - e^2 n/(k_B T))}{R},$$

so equivalently

$$PV = Nk_B T \left(1 - \frac{e^2 n}{k_B T} \right), \quad V = \pi R^2 L.$$

(e) The non-interacting free ions behave as in (a)-(d) with $n \rightarrow n_2$. Stability requires the effective control parameter be at the critical value, hence

$$n_2^* = \frac{k_B T}{e^2} \Rightarrow N_2^* = \frac{k_B T}{e^2} L.$$

If $n_2 > n_2^*$ at $T < T_c$, ions condense to reduce n_2 ; if $n_2 < n_2^*$, ions evaporate to increase n_2 . For $T > T_c$, $n_2^* = n$ and there is no condensation.

JUDGE RATIONALE

- **(a)** Correct partition function, including the thermal wavelength, indistinguishability factor, cylindrical spatial integral, and the logarithmic critical case $q=1$.
- **(b)** Correct normalized radial distribution and first moment, with correct limiting forms at $q=1$ and $q=3/2$.
- **(c)** Correct transition temperature $T_c = e^2 n/k_B$ and correct R/a scaling from the stated mean-radius formula, including the refined subregimes below T_c .
- **(d)** Correct wall-pressure expression from the R derivative of $\ln Z$. It matches the reference above T_c , vanishes asymptotically below T_c , and gives the critical logarithmic limit.
- **(e)** Correct equilibrium free-ion number, including saturation at N for $T > T_c$ and condensation to $N_2^* = L k_B T/e^2$ when below the transition.

5. kinetic/3.5

PROBLEM STATEMENT

Two dimensional electron gas in a magnetic field: Consider a classical gas of electrons confined to two dimensions in a uniform magnetic field $\vec{B}$.

- (a) For non-interacting electrons with Hamiltonian $\mathcal{H} = \sum_i [(\vec{p}_i - e\vec{A})^2/(2m) \pm \mu_B |\vec{B}|]$ with $\vec{A} = (\vec{B} \times \vec{q})/2$, obtain the classical equations of motion.
- (b) Write the Boltzmann equations for the densities $f_{\uparrow}(\mathbf{p}, \mathbf{q}, t)$ and $f_{\downarrow}(\mathbf{p}, \mathbf{q}, t)$ of electrons with up and down spins, in terms of the two cross-sections $\sigma \equiv \sigma_{\uparrow\uparrow} = \sigma_{\downarrow\downarrow}$, and $\sigma_{\times} \equiv \sigma_{\uparrow\downarrow}$ for spin-conserving collisions.
- (g) Write down the most general form for the equilibrium distribution functions for particles confined to a circularly symmetric potential.

EXTRACTED ANSWER

Let $\boldsymbol{\pi} = \mathbf{p} - e\mathbf{A}(\mathbf{q})$, $\mathbf{v} = \boldsymbol{\pi}/m$, $\epsilon_{\uparrow,\downarrow} = \pm\mu_B|\mathbf{B}|$, $\vec{\uparrow} = \downarrow$, $\vec{\downarrow} = \uparrow$.

(a) For each particle, $\dot{\mathbf{q}} = \frac{\mathbf{p} - e\mathbf{A}}{m}$, $\dot{p}_i = e\dot{q}_j \partial_i A_j$, $m\dot{\mathbf{v}} = e\mathbf{v} \times \mathbf{B}$.

For $\mathbf{B} = B\hat{\mathbf{z}}$, $\mathbf{A} = (-By/2, Bx/2)$, this is $\dot{x} = \frac{p_x + eBy/2}{m}$, $\dot{y} = \frac{p_y - eBx/2}{m}$, $\dot{p}_x = \frac{eB}{2}\dot{y}$, $\dot{p}_y = -\frac{eB}{2}\dot{x}$, equivalently $m\ddot{x} = eB\dot{y}$, $m\ddot{y} = -eB\dot{x}$. The constant Zeeman terms ϵ_s do not affect the orbit.

(b) The Boltzmann equations are

$$[\partial_t + \dot{\mathbf{q}} \cdot \nabla_{\mathbf{q}} + \dot{\mathbf{p}} \cdot \nabla_{\mathbf{p}}] f_s(\mathbf{p}, \mathbf{q}, t) = C_s[f_{\uparrow}, f_{\downarrow}], \quad s = \uparrow, \downarrow,$$

$$C_{\uparrow} = \mathcal{I}_{\uparrow\uparrow}^{\sigma} + \mathcal{I}_{\uparrow\downarrow}^{\sigma_{\times}}, \quad C_{\downarrow} = \mathcal{I}_{\downarrow\downarrow}^{\sigma} + \mathcal{I}_{\downarrow\uparrow}^{\sigma_{\times}},$$

$$\mathcal{I}_{sr}^{\kappa}(\mathbf{p}_1) = \int d^2p_2 \int d\theta g \kappa(g, \theta) [f_s(\mathbf{p}'_1) f_r(\mathbf{p}'_2) - f_s(\mathbf{p}_1) f_r(\mathbf{p}_2)],$$

$g = |(\mathbf{p}_1 - e\mathbf{A}) - (\mathbf{p}_2 - e\mathbf{A})|/m$, $\mathbf{p}'_1, \mathbf{p}'_2$ are the pre-collision momenta giving $\mathbf{p}_1, \mathbf{p}_2$ by an elastic spin-conserving collision.

(g) With a circular potential $U(r)$, define $H_s = \frac{(\mathbf{p} - e\mathbf{A})^2}{2m} + U(r) + \epsilon_s$, $L_z = (\mathbf{q} \times \mathbf{p})_z$.

Collisionless stationary distributions may be any $F_s(H_s, L_z)$. For Boltzmann equilibrium with opposite-spin collisions present,

$$f_s^{\text{eq}}(\mathbf{p}, \mathbf{q}) = A_s \exp \left[-\beta \left(\frac{(\mathbf{p} - e\mathbf{A})^2}{2m} + U(r) + \epsilon_s - \Omega L_z \right) \right], \quad s = \uparrow, \downarrow,$$

where $A_{\uparrow}, A_{\downarrow}$ are independent normalizations/chemical potentials, while β and Ω are common to both spin species, with parameters chosen so the distri

FINAL SOLUTION

(a) With $\vec{A} = (\vec{B} \times \vec{q})/2$, Hamilton's equations yield $\dot{\vec{q}} = \partial\mathcal{H}/\partial\vec{p} = (\vec{p} - e\vec{A})/m$, and $\dot{\vec{p}} = -\partial\mathcal{H}/\partial\vec{q} = (e/m)(\vec{p} - e\vec{A}) \times \vec{B}/2 + \dots$. Eliminating $\vec{p}$ gives $m\ddot{\vec{q}} = e\dot{\vec{q}} \times \vec{B}$, i.e. cyclotron motion in planes perpendicular to $\vec{B}$ with $\bar{\omega}_c = (e/m)\vec{B}$.

(b) The Boltzmann equations with elastic spin-conserving collisions are

$$\partial_t f_{\uparrow} - \{\mathcal{H}_{\uparrow}, f_{\uparrow}\} = \int d^2p_2 d\Omega |v_1 - v_2| [(d\sigma/d\Omega)(f'_{\uparrow} f_{\uparrow}^2 - f_{\uparrow} f_{\uparrow}^2) + (d\sigma_{\times}/d\Omega)(f'_{\uparrow} f'_{\downarrow} - f_{\uparrow} f_{\downarrow})], \text{ and similarly for } \downarrow \text{ with } \uparrow \leftrightarrow \downarrow.$$

(c) Define $H = \int d^2q d^2p [f_{\uparrow} \ln f_{\uparrow} + f_{\downarrow} \ln f_{\downarrow}]$. Using the standard symmetrization of the collision integrals, one finds each contribution is non-positive, hence $dH/dt \leq 0$.

- (d) If at each $\vec{q}$, $\ln f_\alpha(\vec{p})$ is a linear combination of collision invariants, then for every binary collision the sums of $\ln f$ before and after are equal, making each collision contribution vanish; thus $dH/dt = 0$.
- (e) If $f_\alpha = f_\alpha(I_1, I_2, \dots)$ depends only on invariants I_j with $\{I_j, \mathcal{H}_\alpha\} = 0$, then the streaming term vanishes: $\{f_\alpha, \mathcal{H}_\alpha\} = \sum_j (\partial f_\alpha / \partial I_j) \{I_j, \mathcal{H}_\alpha\} = 0$.
- (f) During collisions at fixed $\vec{q}$, momentum conservation implies $\vec{q} \times (\vec{p}_1 + \vec{p}_2) = \vec{q} \times (\vec{p}_1' + \vec{p}_2')$ so $\vec{L}$ is conserved. Away from collisions, for the given $\mathcal{H}$, $\{\mathcal{H}, \vec{q} \times \vec{p}\} = 0$; in 2D the conserved component is L_z .
- (g) The most general equilibrium form compatible with (e) and collisions is $f_\alpha = A_\alpha \exp[-\beta \mathcal{H}_\alpha - \gamma L_z]$ (possible rigid rotation via γ).
- (h) Impurities break momentum (and L_z) conservation, removing the γ term. Magnetic impurities mix spins implying $A_\uparrow = A_\downarrow$; non-magnetic do not.
- (i) Angular momentum conservation does not yield an independent hydrodynamic equation beyond momentum; spin conservation yields a new equation for magnetization (spin density difference).

JUDGE RATIONALE

- **(a)** Correct Hamilton equations are given, including the Lorentz-force form after eliminating canonical momentum and the fact that the constant Zeeman term does not affect the orbital motion.
- **(b)** The Boltzmann equations include the correct streaming term and collision integrals for same-spin and opposite-spin elastic spin-conserving collisions with the two cross sections, in an equivalent notation.
- **(g)** The equilibrium distribution is given as a Maxwell-Boltzmann exponential with common beta and rigid-rotation angular momentum term, with spin-dependent normalizations, matching the reference form up to the equivalent convention $\gamma = -\beta \Omega$ and including the circular potential.

6. probability/2.5

PROBLEM STATEMENT

5. Benford's law describes the observed probabilities of the first digit in a great variety of data sets, such as stock prices. Rather counter-intuitively, the digits 1 through 9 occur with probabilities 0.301, 0.176, 0.125, 0.097, 0.079, 0.067, 0.058, 0.051, 0.046 respectively. The key observation is that this distribution is invariant under a change of scale. Find a formula that fits the above probabilities on the basis of this observation.

EXTRACTED ANSWER

For first digit $d = 1, \dots, 9$, $P(d) = \log_{10} \left(\frac{d+1}{d} \right) = \log_{10} \left(1 + \frac{1}{d} \right)$, giving (0.3010, 0.1761, 0.1249, 0.0969, 0.0792, 0.0669, 0.0580, 0.0512, 0.0458) for $d = 1, \dots, 9$.

FINAL SOLUTION

Consider products $x = \prod_{i=1}^N r_i$ with positive random r_i . Then $\ell = \ln x = \sum \ln r_i$ is Gaussian for large N (CLT), so x is log-normal: $p(x) \propto x^{-1} \exp[-(\ln x - \mu)^2 / (2\sigma^2)]$. The probability that the first digit is i is

$$p_i \approx \sum_q \int_{10^q}^{10^{q+1}} p(x) dx = \sum_q \int_{q+\ln i}^{q+\ln(i+1)} p(\ell) d\ell \propto \ln(i+1) - \ln i = \ln \left(1 + \frac{1}{i} \right).$$

Normalizing in base 10 yields Benford's law

$$p_i = \log_{10} \left(1 + \frac{1}{i} \right), \quad i = 1, \dots, 9,$$

which matches the listed frequencies.

JUDGE RATIONALE

- **(a)** The candidate gives the Benford formula $P(d) = \log_{10}((d+1)/d) = \log_{10}(1 + 1/d)$ for $d = 1, \dots, 9$ and the corresponding probabilities, matching the reference solution.

7. textbook/3.8.9

PROBLEM STATEMENT

3.8.9 Pauli paramagnetism

Consider electrons with Zeeman coupling $\hat{H}_z = -\mu_0 \mathbf{B} \cdot \hat{\mathbf{S}}$, $\hat{\mathbf{S}} = \frac{1}{2} a_{\alpha\sigma}^\dagger \boldsymbol{\sigma}_{\sigma\sigma'} a_{\alpha\sigma'}$, and a non-magnetic part $\hat{H}_0 = \sum_{\alpha,\sigma} a_{\alpha\sigma}^\dagger \epsilon_\alpha a_{\alpha\sigma}$. A magnetic field lifts spin degeneracy near the Fermi surface and produces a Pauli paramagnetic response.

(a) Construct the coherent-state action of $\hat{H} = \hat{H}_0 + \hat{H}_z$. Integrate out the Grassmann fields to obtain the free energy F as a sum over Matsubara frequencies.

(b) Determine the low-temperature spin contribution to the susceptibility $\chi \equiv -\partial_B^2 F|_{B=0}$, where $\rho(\epsilon) = \sum_\alpha \delta(\epsilon - \epsilon_\alpha)$ is the single-particle density of states.

EXTRACTED ANSWER

$$\begin{aligned}
 \text{(a)} \quad S &= \int_0^\beta d\tau \bar{\psi}_{\alpha\sigma} \left[(\partial_\tau + \epsilon_\alpha - \mu) \delta_{\sigma\sigma'} - \frac{\mu_0}{2} \mathbf{B} \cdot \boldsymbol{\sigma}_{\sigma\sigma'} \right] \psi_{\alpha\sigma'}, \\
 F &= -T \sum_{\alpha,n} \ln \det \left[(-i\omega_n + \epsilon_\alpha - \mu) \mathbf{1} - \frac{\mu_0}{2} \mathbf{B} \cdot \boldsymbol{\sigma} \right] \\
 &= -T \sum_{\alpha,n} \ln \left[(-i\omega_n + \epsilon_\alpha - \mu)^2 - \left(\frac{\mu_0 B}{2} \right)^2 \right] \quad \text{up to a } B\text{-independent constant;} \\
 \text{(b)} \quad \chi &= \frac{\mu_0^2}{2} \int d\epsilon \rho(\epsilon) \left[-\frac{\partial n_F(\epsilon - \mu)}{\partial \epsilon} \right] = \frac{\mu_0^2}{2} \left[\rho(\mu) + \frac{\pi^2 T^2}{6} \rho''(\mu) + O(T^4) \right], \\
 \text{hence at } T &= 0, \quad \chi = \frac{\mu_0^2}{2} \rho(\epsilon_F).
 \end{aligned}$$

FINAL SOLUTION

(a) Choosing the field along the quantization axis, $\hat{H} = \sum_{\alpha\sigma} a_{\alpha\sigma}^\dagger (\epsilon_\alpha - \frac{\mu_0 B}{2} (\sigma_z)_{\sigma\sigma}) a_{\alpha\sigma}$. The action in frequency space is $S = \sum_{\alpha\sigma n} \bar{\psi}_{\alpha\sigma n} (-i\omega_n + \xi_\alpha - \frac{\mu_0 B}{2} (\sigma_z)_{\sigma\sigma}) \psi_{\alpha\sigma n}$. Integrating out ψ gives

$$F = -T \sum_{\alpha n} \ln \left(\beta^2 [(-i\omega_n + \xi_\alpha)^2 - \frac{1}{4} (\mu_0 B)^2] \right).$$

(b) Differentiating twice and setting $B = 0$ yields $\chi = -\frac{1}{2} \mu_0^2 T \sum_{\alpha n} (-i\omega_n + \xi_\alpha)^{-2}$. Performing the Matsubara sum using residues gives $\chi = -(\mu_0^2/2) \sum_\alpha n'_F(\xi_\alpha) = -(\mu_0^2/2) \int d\epsilon \rho(\epsilon) n'_F(\epsilon - \mu)$. As $T \rightarrow 0$, $n'_F(\epsilon - \mu) \rightarrow -\delta(\epsilon - \mu)$, hence $\chi \rightarrow (\mu_0^2/2) \rho(\mu)$.

JUDGE RATIONALE

- **(a)** Correctly gives the coherent-state action with chemical potential included as $\xi = \epsilon_\alpha - \mu$, includes the Zeeman matrix term, and integrates out the Grassmann fields to obtain the Matsubara log determinant/free energy. The missing beta factor is explicitly only a B-independent constant, consistent with the reference.

- **(b)** Correctly obtains the spin susceptibility as $(\mu_0^2/2) \int \rho(\epsilon) (-\partial n_F / \partial \epsilon)$, and gives the $T=0$ limit $(\mu_0^2/2) \rho(\mu)$. The Sommerfeld correction is consistent and does not introduce a material error.

8. textbook/5.4

PROBLEM STATEMENT

Two dimensional Coulomb gas: Consider a classical mixture of N positive and N negative charged particles in a two dimensional box of area $A = L \times L$. The Hamiltonian is

$$\mathcal{H} = \sum_{i=1}^{2N} \frac{\vec{p}_i^2}{2m} - \sum_{i<j} c_i c_j \ln |\vec{q}_i - \vec{q}_j|,$$

where $c_i = +c_0$ for $i = 1, \dots, N$ and $c_i = -c_0$ for $i = N+1, \dots, 2N$.

(a) In the interaction term each pair appears only once, and there is no self interaction. How many pairs have repulsive interactions, and how many have attractive interactions?

(b) Write down the partition function $Z(N, T, A)$ as an integral over $\{\vec{q}_i\}, \{\vec{p}_i\}$. Perform the momentum integrals and rewrite the coordinate contribution as a product with powers of $|\vec{q}_i - \vec{q}_j|$.

(c) Although the integrals over $\{\vec{q}_i\}$ are not feasible exactly, the dependence of Z on A can be obtained by rescaling $\vec{q}_i = \vec{q}_i/L$. Using results from (a) and (b), give the dependence $Z \propto A^{2N - \beta c_0^2 N/2}$.

(d) Calculate the two-dimensional pressure P of this gas.

(e) A hard core forbids any two particles from approaching closer than a . By examining Z for $N = 1$ (two opposite charges), estimate the temperature T_c at which the short-distance scale a becomes important and the result of (d) fails.

EXTRACTED ANSWER

(a) $N(N-1)$ repulsive pairs, N^2 attractive pairs.

$$\begin{aligned} \text{(b)} \quad Z(N, T, A) &= \frac{1}{(N!)^2 h^{4N}} \int \prod_{i=1}^{2N} d^2 q_i d^2 p_i e^{-\beta \mathcal{H}} \\ &= \frac{1}{(N!)^2 h^{4N}} \left(\frac{2\pi m}{\beta} \right)^{2N} \int_A \prod_{i=1}^{2N} d^2 q_i \prod_{i<j} |\vec{q}_i - \vec{q}_j|^{\beta c_i c_j} \\ &= \frac{1}{(N!)^2 h^{4N}} \left(\frac{2\pi m}{\beta} \right)^{2N} \int_A \prod_{i=1}^{2N} d^2 q_i \frac{\prod_{i<j} |\vec{q}_i - \vec{q}_j|^{\beta c_0^2}}{\prod_{\substack{i,j \\ c_i = +c_0, c_j = -c_0}} |\vec{q}_i - \vec{q}_j|^{\beta c_0^2}}. \end{aligned}$$

(c) $Z \propto A^{2N - \beta c_0^2 N/2}$.

$$\text{(d)} \quad P = k_B T \left(\frac{\partial \ln Z}{\partial A} \right)_{N,T} = \frac{1}{A} \left(2N k_B T - \frac{N c_0^2}{2} \right).$$

(e) $T_c = \frac{c_0^2}{2k_B}$; for $T \leq T_c$ the hard-core scale a controls the integral and (d) fails.

FINAL SOLUTION

(a) There are N positive and N negative charges. Opposite-charge pairs: $N \times N = N^2$ (attractive). Like-charge pairs: choose 2 among N positives or N negatives: $2 \binom{N}{2} = N(N-1)$ (repulsive).

(b) The partition function is

$$Z(N, T, A) = \frac{1}{(N!)^2 h^{4N}} \int \prod_{i=1}^{2N} d^2 \vec{q}_i d^2 \vec{p}_i \exp \left[-\beta \sum_{i=1}^{2N} \frac{p_i^2}{2m} + \beta \sum_{i<j} c_i c_j \ln |\vec{q}_i - \vec{q}_j| \right].$$

Gaussian momentum integrals give $\lambda = h/\sqrt{2\pi m k_B T}$ and

$$Z(N, T, A) = \frac{1}{\lambda^{4N} (N!)^2} \int \prod_{i=1}^{2N} d^2 \vec{q}_i \prod_{i < j}^{2N} |\vec{q}_i - \vec{q}_j|^{\beta c_i c_j}.$$

(c) With $\vec{q}_i = L \vec{q}'_i$, one has $d^2 \vec{q}_i = L^2 d^2 \vec{q}'_i$ and $|\vec{q}_i - \vec{q}_j| = L |\vec{q}'_i - \vec{q}'_j|$. Counting powers of L , the measure contributes L^{4N} ; the interactions contribute $L^{\beta c_0^2 [N(N-1)]}$ from repulsions and $L^{-\beta c_0^2 [N^2]}$ from attractions. Hence

$$Z(N, T, A) \propto L^{4N - \beta c_0^2 N} = A^{2N - (\beta c_0^2 N)/2}.$$

(d) The pressure is

$$P = k_B T \left. \frac{\partial \ln Z}{\partial A} \right|_{N, T} = k_B T \frac{\partial}{\partial A} \left[\left(2N - \frac{\beta c_0^2 N}{2} \right) \ln A + \text{const} \right] = \frac{2N k_B T}{A} - \frac{N c_0^2 k_B T}{2A}.$$

At high T this reduces to the ideal-gas value for $2N$ particles; at low T the expression becomes negative, signaling the breakdown due to collapse.

(e) For one positive and one negative charge ($N = 1$), with hard core a and system size L , the coordinate integral in center-of-mass and relative variables is

$$Z(N=1, T, A) = \frac{A}{\lambda^4} \int d^2 \vec{q} q^{-\beta c_0^2} \approx \frac{2\pi A}{\lambda^4} \int_a^L dq q^{1-\beta c_0^2} = \frac{2\pi A}{\lambda^4} \frac{L^{2-\beta c_0^2} - a^{2-\beta c_0^2}}{2 - \beta c_0^2}.$$

If $2 - \beta c_0^2 < 0$ (low T), the short-distance cutoff a dominates; if $2 - \beta c_0^2 > 0$ (high T), the system size L dominates as in (c). The crossover is at $\beta c_0^2 = 2$, i.e.

$$T_c = \frac{c_0^2}{2k_B}.$$

For $T > T_c$ the system is a dissociated plasma; for $T < T_c$ it forms tightly bound opposite-charge dipoles (no collapse due to the hard core).

JUDGE RATIONALE

- **(a)** Correctly gives $N(N-1)$ like-charge repulsive pairs and N^2 opposite-charge attractive pairs.
- **(b)** Correctly writes the indistinguishable-particle partition function, performs the Gaussian momentum integrals, and expresses the coordinate integral as products with powers $\beta c_i c_j$, including like-charge factors in the numerator and opposite-charge factors in the denominator.
- **(c)** Correctly states the area dependence Z proportional to $A^{(2N - \beta c_0^2 N/2)}$.
- **(d)** Correctly differentiates the area dependence to obtain $P = (2N k_B T - N c_0^2/2)/A$, equivalent to $k_B T(2N - \beta c_0^2 N/2)/A$.

- **(e)** Correctly identifies the crossover $\beta c_0^2 = 2$, giving $T_c = c_0^2/(2 k_B)$, and states that below this scale the hard core controls the integral and the pressure result fails.

9. lattice/6.6

PROBLEM STATEMENT

One dimensional gas: The transfer matrix method can also be applied to a one dimensional gas of particles with short-range interactions. For a potential with a hard core that screens the interactions from further neighbors, the Hamiltonian for N particles is

$$\mathcal{H} = \sum_{i=1}^N \frac{p_i^2}{2m} + \sum_{i=2}^N \mathcal{V}(x_i - x_{i-1}).$$

The (indistinguishable) particles are labeled with coordinates $\{x_i\}$ such that

$$0 \leq x_1 \leq x_2 \leq \dots \leq x_N \leq L,$$

where L is the length of the box confining the particles.

(b) Write the expression for the partition function $Z(T, N, L)$. Change variables to $\delta_1 = x_1, \delta_2 = x_2 - x_1, \dots, \delta_N = x_N - x_{N-1}$, and carefully indicate the allowed ranges of integration and the constraints.

(c) Consider the Gibbs partition function obtained from the Laplace transformation

$$Z(T, N, P) = \int_0^\infty dL \exp(-\beta PL) Z(T, N, L),$$

and find the standard formula for P in the canonical ensemble.

(d) Change variables from L to $\delta_{N+1} = L - \sum_{i=1}^N \delta_i$, and find the expression for $Z(T, N, P)$ as a product over one-dimensional integrals over each δ_i .

(e) At a fixed pressure P , find expressions for the mean length $L(T, N, P)$, and the density $n = N/L(T, N, P)$ (involving ratios of integrals).

(f) For a hard sphere gas, with minimum separation a between particles, calculate the equation of state $P(T, n)$.

EXTRACTED ANSWER

$$(b) \quad Z(T, N, L) = \lambda_T^{-N} \int_0^\infty \prod_{i=1}^N d\delta_i \Theta\left(L - \sum_{i=1}^N \delta_i\right) \exp\left[-\beta \sum_{i=2}^N \mathcal{V}(\delta_i)\right],$$

$$\delta_1 = x_1, \quad \delta_i = x_i - x_{i-1} \quad (2 \leq i \leq N), \quad \delta_i \geq 0, \quad \sum_{i=1}^N \delta_i \leq L.$$

$$(c) \quad Z(T, N, P) = \int_0^\infty dL e^{-\beta PL} Z(T, N, L), \quad P = k_B T \left(\frac{\partial \ln Z(T, N, L)}{\partial L} \right)_{T, N}.$$

$$(d) \quad \delta_{N+1} = L - \sum_{i=1}^N \delta_i \geq 0, \quad Z(T, N, P) = \lambda_T^{-N} \left(\int_0^\infty d\delta e^{-\beta P\delta} \right)^2 \left(\int_0^\infty d\delta e^{-\beta[P\delta + \mathcal{V}(\delta)]} \right)^{N-1}.$$

$$(e) \quad L(T, N, P) = \frac{2}{\beta P} + (N-1) \frac{\int_0^\infty d\delta \delta e^{-\beta[P\delta + \mathcal{V}(\delta)]}}{\int_0^\infty d\delta e^{-\beta[P\delta + \mathcal{V}(\delta)]}}, \quad n = \frac{N}{L(T, N, P)}.$$

$$(f) \quad \mathcal{V}(\delta) = \infty \text{ for } \delta < a, \quad \mathcal{V}(\delta) = 0 \text{ for } \delta \geq a, \\ L = \frac{2}{\beta P} + (N-1) \left(a + \frac{1}{\beta P} \right), \quad \text{and in the thermodynamic limit} \quad P = \frac{nk_B T}{1 - na}.$$

FINAL SOLUTION

(a) Hard cores screen further-neighbor interactions, leaving only nearest-neighbor potentials. Including kinetic energies,

$$\mathcal{H} = \sum_{i=1}^N \frac{p_i^2}{2m} + \sum_{i=2}^N \mathcal{V}(x_i - x_{i-1}), \quad 0 \leq x_1 \leq x_2 \leq \dots \leq x_N \leq L.$$

(b) The canonical partition function is

$$Z(T, N, L) = \frac{1}{h^N} \int_0^L dx_1 \int_{x_1}^L dx_2 \cdots \int_{x_{N-1}}^L dx_N e^{-\beta \sum_{i=2}^N \mathcal{V}(x_i - x_{i-1})} \prod_{i=1}^N \int_{-\infty}^{\infty} dp_i e^{-\beta p_i^2 / (2m)}.$$

After Gaussian momentum integrals with $\lambda = h / \sqrt{2\pi m k_B T}$,

$$Z(T, N, L) = \frac{1}{\lambda^N} \int_0^L dx_1 \int_{x_1}^L dx_2 \cdots \int_{x_{N-1}}^L dx_N e^{-\beta \sum_{i=2}^N \mathcal{V}(x_i - x_{i-1})}.$$

Change variables $\delta_1 = x_1, \delta_i = x_i - x_{i-1} (i \geq 2)$ so that $x_N = \sum_{i=1}^N \delta_i$. Then

$$Z(T, N, L) = \frac{1}{\lambda^N} \int_0^L d\delta_1 \int_0^{L-\delta_1} d\delta_2 \cdots \int_0^{L-\sum_{i=1}^{N-1} \delta_i} d\delta_N e^{-\beta \sum_{i=2}^N \mathcal{V}(\delta_i)}.$$

Equivalently,

$$Z(T, N, L) = \frac{1}{\lambda^N} \int_0^\infty \prod_{i=1}^N d\delta_i e^{-\beta \sum_{i=2}^N \mathcal{V}(\delta_i)} \Theta\left(L - \sum_{i=1}^N \delta_i\right).$$

(c) The Gibbs partition function is

$$Z(T, N, P) = \int_0^\infty dL e^{-\beta PL} Z(T, N, L).$$

Extremizing the integrand with respect to L gives the canonical pressure relation

$$P = k_B T \left. \frac{\partial \ln Z}{\partial L} \right|_{T, N},$$

consistent with $F = -k_B T \ln Z$ and $P = -\partial F / \partial L|_{T, N}$.

(d) Performing the Laplace transform using the step-function representation yields

$$Z(T, N, P) = \frac{1}{\lambda^N (\beta P)^2} \left\{ \int_0^\infty d\delta e^{-\beta[\mathcal{V}(\delta) + P\delta]} \right\}^{N-1}.$$

An equivalent derivation introduces $\delta_{N+1} = L - \sum_{i=1}^N \delta_i$ and integrates each δ_i independently from 0 to ∞ .

(e) The mean length is

$$L(T, N, P) = -k_B T \left. \frac{\partial}{\partial P} \ln Z(T, N, P) \right|_{T, N} = \frac{2}{\beta P} + (N-1) \frac{\int_0^\infty d\delta \delta e^{-\beta[\mathcal{V}(\delta)+P\delta]}}{\int_0^\infty d\delta e^{-\beta[\mathcal{V}(\delta)+P\delta]}}.$$

Thus

$$n = \frac{N}{L(T, N, P)} = N \left\{ \frac{2k_B T}{P} + (N-1) \frac{\int_0^\infty d\delta \delta e^{-\beta[\mathcal{V}(\delta)+P\delta]}}{\int_0^\infty d\delta e^{-\beta[\mathcal{V}(\delta)+P\delta]}} \right\}^{-1}.$$

For an ideal gas ($\mathcal{V} = 0$), $L_{i.g.}(T, N, P) = (N+1)k_B T/P$, and $n_{i.g.} = \frac{N}{N+1} \frac{P}{k_B T}$.

(f) Hard spheres: impose $\delta_i \geq a$ for $i = 2, \dots, N$, so

$$Z(T, N, P) = \frac{1}{\lambda^N (\beta P)^2} \left[\int_a^\infty d\delta e^{-\beta P \delta} \right]^{N-1} = \frac{1}{\lambda^N} \left(\frac{1}{\beta P} \right)^{N+1} e^{-\beta P a (N-1)}.$$

Then

$$L = -k_B T \left. \frac{\partial \ln Z}{\partial P} \right|_{T, N} = \frac{(N+1)k_B T}{P} + (N-1)a.$$

Rearranging,

$$\beta P = \frac{N+1}{L - (N-1)a} = \frac{n + 1/L}{1 - (n-1/L)a} \approx n(1 + na + n^2 a^2 + \dots),$$

so the virial series is $\beta P \approx n + an^2 + a^2 n^3 + \dots$ with $B_\ell(T) = a^{\ell-1}$.

JUDGE RATIONALE

- **(b)** Correctly gives the canonical partition function after momentum integration, changes to the delta variables, and states the nonnegative ranges with the constraint $\sum_i \delta_i \leq L$.
- **(c)** Correctly states the Laplace-transformed Gibbs partition function and the canonical pressure relation $P = k_B T (\partial \ln Z / \partial L)_{T, N}$.
- **(d)** Correctly introduces δ_{N+1} and factorizes the Gibbs partition function into two free-gap integrals and $N-1$ interacting nearest-neighbor integrals, equivalent to the reference expression.
- **(e)** Correctly gives the mean length as $2/(\beta P)$ plus $(N-1)$ times the ratio of first-moment and zeroth-moment interaction integrals, and states $n = N/L$.
- **(f)** Correctly implements the hard-core lower limit, obtains $L = (N+1)/(\beta P) + (N-1)a$, and gives the thermodynamic-limit equation of state $P = n k_B T/(1-na)$.

10. textbook/7.13

PROBLEM STATEMENT

Freezing of He⁴: At low temperatures He⁴ can be converted from liquid to solid by application of pressure. An interesting feature of the phase boundary is that the melting pressure is reduced slightly from its $T = 0$ K value, by approximately 20 N m^{-2} at its minimum at $T = 0.8$ K. We will use a simple model of liquid and solid phases of He⁴ to account for this feature.

(a) The important excitations in liquid He⁴ at $T < 1$ K are phonons of velocity c . Calculate the contribution of these modes to the heat capacity per particle C_V^ℓ/N , of the liquid.

(b) Calculate the low temperature heat capacity per particle C_V^s/N , of solid He⁴ in terms of longitudinal and transverse sound velocities c_L , and c_T .

(c) Using the above results calculate the entropy difference $(s_\ell - s_s)$, assuming a single sound velocity $c \approx c_L \approx c_T$, and approximately equal volumes per particle $v_\ell \approx v_s \approx v$. Which phase (solid or liquid) has the higher entropy?

(d) Assuming a small (temperature independent) volume difference $\delta v = v_\ell - v_s$, calculate the form of the melting curve. To explain the anomaly described at the beginning, which phase (solid or liquid) must have the higher density?

EXTRACTED ANSWER

$$\begin{aligned}
 \text{(a)} \quad \frac{C_V^\ell}{N} &= \frac{2\pi^2}{15} v_\ell \frac{k_B^4 T^3}{\hbar^3 c^3} = \frac{2\pi^2}{15} k_B v_\ell \left(\frac{k_B T}{\hbar c} \right)^3. \\
 \text{(b)} \quad \frac{C_V^s}{N} &= \frac{2\pi^2}{15} v_s \frac{k_B^4 T^3}{\hbar^3} \left(\frac{1}{c_L^3} + \frac{2}{c_T^3} \right). \\
 \text{(c)} \quad s_\ell - s_s &= \frac{2\pi^2}{45} v \frac{k_B^4 T^3}{\hbar^3} \left(\frac{1}{c^3} - \frac{3}{c^3} \right) = -\frac{4\pi^2}{45} v \frac{k_B^4 T^3}{\hbar^3 c^3}, \quad \text{so the solid has higher entropy.} \\
 \text{(d)} \quad \frac{dP_m}{dT} &= \frac{s_\ell - s_s}{\delta v} = -\frac{4\pi^2}{45} v \frac{k_B^4 T^3}{\hbar^3 c^3}, \\
 P_m(T) - P_m(0) &= -\frac{\pi^2}{45} v \frac{k_B^4 T^4}{\hbar^3 c^3}. \quad \text{For the melting pressure to decrease with } T, \delta v > 0, \text{ so } v_\ell > v_s; \text{ the solid must have the higher density.}
 \end{aligned}$$

FINAL SOLUTION

$$\text{(a) For phonons with } \varepsilon = \hbar ck, E_{\text{ph}} = \sum_{\vec{k}} \hbar ck / (e^{\beta \hbar ck} - 1) = \frac{V}{2\pi^2} \hbar c \left(\frac{k_B T}{\hbar c} \right)^4 \int_0^\infty dx \frac{x^3}{e^x - 1} = \frac{\pi^2}{30} V \hbar c (k_B T / \hbar c)^4.$$

Hence

$$C_V = dE/dT = \frac{2\pi^2}{15} V k_B (k_B T / \hbar c)^3, \text{ so } C_V^\ell / N = \frac{2\pi^2}{15} k_B v_\ell (k_B T / \hbar c)^3.$$

(b) In the solid there are two transverse modes (c_T) and one longitudinal (c_L), yielding

$$C_V^s / N = \frac{2\pi^2}{15} k_B v_s (k_B T / \hbar)^3 (2/c_T^3 + 1/c_L^3).$$

(c) The entropies are $s = \int_0^T C_V(T') dT' / T'$. Thus

$$s_\ell = \frac{2\pi^2}{45} k_B v_\ell (k_B T / \hbar c)^3, \quad s_s = \frac{2\pi^2}{45} k_B v_s (k_B T / \hbar)^3 (2/c_T^3 + 1/c_L^3). \quad \text{For } c_L \approx c_T \approx c \text{ and } v_\ell \approx v_s \approx v, s_\ell - s_s \approx$$

$-\frac{4\pi^2}{45}k_B v (k_B T/\hbar c)^3$. The solid has higher entropy (three phonon branches vs one).

(d) Clausius-Clapeyron gives $dP/dT = (s_\ell - s_s)/(v_\ell - v_s)$. With $\delta v = v_\ell - v_s > 0$ to explain decreasing melting pressure, one obtains

$$\frac{dP}{dT} = -\frac{4\pi^2}{45}k_B \frac{v}{\delta v} (k_B T/\hbar c)^3, \text{ integrating to}$$

$$P_{\text{melt}}(T) = P(0) - \frac{\pi^2}{45}k_B \frac{v}{\delta v} (k_B T/\hbar c)^3 T. \text{ Thus the solid must have higher density.}$$

JUDGE RATIONALE

- **(a)** Correct heat capacity per particle for one phonon branch in the liquid, equivalent to the reference expression.
- **(b)** Correct solid heat capacity per particle with one longitudinal and two transverse phonon branches.
- **(c)** Correct entropy difference under the equal-velocity and equal-volume assumptions, and correctly identifies the solid as having higher entropy.
- **(d)** Correct Clausius-Clapeyron form and integrated melting curve, and correctly states that decreasing melting pressure requires $\delta v > 0$, so the solid has higher density.

11. optics/2-58

PROBLEM STATEMENT

A laser beam ($\lambda = 6.3 \times 10^{-5} \text{ cm}$) is directed at grazing incidence onto a steel machinist's ruler (which is calibrated in $1/16 \text{ inch}$). The light reflects from the surface of the ruler and is projected onto a vertical wall 10 meters away (Fig. 2.61).

(a) Derive the condition on θ' for interference maxima on the wall. Assume for simplicity that the laser beam is initially almost parallel to the ruler surface.

(b) What is the vertical separation on the wall of the zeroth and first order spots of the pattern?

EXTRACTED ANSWER

$$(a) d(1 - \cos \theta') = m\lambda, \quad d = \frac{1}{16} \text{ in}, \quad m = 0, 1, 2, \dots; \quad (b) y_1 - y_0 = L \tan \theta'_1 \approx 0.28 \text{ m} = 28 \text{ cm}.$$

FINAL SOLUTION

- (a) The optical path length difference (OPD) of the parallel light rays shown in Fig. 2.61 is

$$\Delta = d(\cos \theta - \cos \theta') \approx d(1 - \cos \theta')$$

as $\theta \approx 0$. Interference maxima occur when $\Delta = m\lambda$ (m is an integer), or

$$\cos \theta' = 1 - m\lambda/d.$$

- (b) When $m = 0$, $\theta' = 0$. When $m = 1$, $\theta'_1 = 1 - \lambda/d = 0.9996$ taking $d = 1$ division ($\frac{1}{16} \text{ inch}$). The vertical separation on the wall between the zeroth and first order spots is

$$x = l \cdot \tan \theta'_1 = 0.28 \text{ m}.$$

JUDGE RATIONALE

- **(a)** Correctly states the interference maximum condition for grazing incidence as $d(1 - \cos \theta') = m\lambda$, equivalent to $\cos \theta' = 1 - m\lambda/d$.

- **(b)** Correctly gives the wall separation as $L \tan(\theta'_1)$ and the numerical value approximately 0.28 m (28 cm), consistent with the reference solution.

12. electro/1_17

PROBLEM STATEMENT

The inside of a grounded spherical metal shell (inner radius R_1 and outer radius R_2) is filled with space charge of uniform charge density ρ .

- (a) Find the electrostatic energy of the system.
 (b) Find the potential at the center.

EXTRACTED ANSWER

$$(a) U = \frac{2\pi\rho^2 R_1^5}{45\epsilon_0}, \quad (b) V(0) = \frac{\rho R_1^2}{6\epsilon_0}, \text{ for space charge filling the cavity } r < R_1.$$

FINAL SOLUTION

Solution:

Consider a concentric spherical surface of radius r ($r < R_1$). Using Gauss' law we get

$$\mathbf{E} = \frac{r\rho}{3\epsilon_0} \mathbf{e}_r.$$

As the shell is grounded, $\varphi(R_1) = 0$, $E = 0$ ($r > R_2$). Thus

$$\varphi(r) = \int_r^{R_1} E dr = \frac{\rho}{6\epsilon_0} (R_1^2 - r^2).$$

The potential at the center is

$$\varphi(0) = \frac{1}{6\epsilon_0} \rho R_1^2.$$

The electrostatic energy is

$$W = \int \frac{1}{2} \rho \varphi dV = \frac{1}{2} \int_0^{R_1} \frac{\rho}{6\epsilon_0} (R_1^2 - r^2) \cdot \rho \cdot 4\pi r^2 dr = \frac{2\rho^2 R_1^5}{45\epsilon_0}.$$

JUDGE RATIONALE

- **(a)** Correct electrostatic energy for the uniformly charged cavity of the grounded shell, matching the evaluated energy integral.

- **(b)** Correct center potential, $V(0) = \rho R_1^2 / (6 \epsilon_0)$.

13. electro/1_72

PROBLEM STATEMENT

Take a very long cylinder of radius r made of insulating material. Spray a cloud of electrons on the surface. They are free to move about the surface so that, at first, they spread out evenly with a charge per unit area σ_0 . Then put the cylinder in a uniform applied electric field perpendicular to the axis of the cylinder. You are asked to think about the charge density on the surface of the cylinder, $\sigma(\theta)$, as a function of the applied electric field E_a . In doing this you may neglect the electric polarizability of the insulating cylinder.

(b) Calculate the solution for

$\sigma(\theta)$ in the case of a conducting cylinder and state the range of value of E_a for which this solution is applicable to the case described here.

EXTRACTED ANSWER

(b) $\sigma(\theta) = \sigma_0 + 2\epsilon_0 E_a \cos \theta$, θ measured from the applied-field direction. This applies to the electron-coated insulating cylinder only while no positive σ

FINAL SOLUTION

Solution:

Use cylindrical coordinates (ρ, θ, z) with the z -axis along the cylindrical axis and the direction of the applied field given by $\theta = 0$. Let the potentials inside and outside the cylinder be φ_I and φ_{II} respectively. As a long cylinder is assumed, φ_I and φ_{II} are independent of z . As there is no charge inside and outside the cylinder, Laplace's equation applies: $\nabla^2 \varphi_I = \nabla^2 \varphi_{II} = 0$. The boundary conditions are

$$\varphi_I|_{\rho=r} = \varphi_{II}|_{\rho=r}, \quad \left(\frac{\partial \varphi_I}{\partial \theta} \right) \Big|_{\rho=r} = \left(\frac{\partial \varphi_{II}}{\partial \theta} \right) \Big|_{\rho=r},$$

$$\begin{aligned} \varphi_I|_{\rho \rightarrow 0} & \text{ is finite,} \\ \varphi_{II}|_{\rho \rightarrow \infty} & \rightarrow -E_a \rho \cos \theta. \end{aligned}$$

Note the first two conditions arise from the continuity of the tangential component of the electric intensity vector. Furthermore as the electrons are free to move about the cylindrical surface, $E_\theta = 0$ on the surface at equilibrium. As z is not involved, try solutions of the form

$$\varphi_I = A_1 + B_1 \ln \rho + C_1 \rho \cos \theta + \frac{D_1}{\rho} \cos \theta,$$

$$\varphi_{II} = A_2 + B_2 \ln \rho + C_2 \rho \cos \theta + \frac{D_2}{\rho} \cos \theta.$$

Then the above boundary conditions require that

$$B_1 = D_1 = 0, \quad C_2 = -E_a,$$

and

$$A_1 + C_1 r \cos \theta = A_2 + B_2 \ln r - E_a r \cos \theta + \frac{D_2}{r} \cos \theta,$$

$$-C_1 \sin \theta = E_a \sin \theta - \frac{D_2}{r^2} \sin \theta = 0.$$

The last equation gives

$$C_1 = 0, \quad D_2 = E_a r^2.$$

Applying Gauss' law to unit length of the cylinder:

$$\oint \mathbf{E}_{II} \cdot d\mathbf{S} = \frac{q}{\varepsilon_0},$$

i.e.,

$$-\frac{B_2}{r} \cdot 2\pi r = \frac{1}{\varepsilon_0} \sigma_0 \cdot 2\pi r,$$

we obtain

$$B_2 = -\frac{\sigma_0 r}{\varepsilon_0}.$$

Neglecting any possible constant potential, we take $A_2 = 0$. Then

$$A_1 = B_2 \ln r = -\frac{\sigma_0 a \ln r}{\varepsilon_0}.$$

We ultimately obtain the following expressions

$$\varphi_I = -\frac{\sigma_0 r \ln r}{\varepsilon_0}, \quad E_I = 0,$$

$$\varphi_{II} = -\frac{\sigma_0 r \ln \rho}{\varepsilon_0} - E_a \rho \cos \theta + \frac{E_a r^2}{\rho} \cos \theta,$$

$$\mathbf{E}_{II} = \left(\frac{\sigma r}{\varepsilon_0 \rho} + E_a \cos \theta + \frac{E_a r^2}{\rho^2} \cos \theta \right) \mathbf{e}_\rho$$

$$-E_a \left(1 - \frac{r^2}{\rho^2} \right) \sin \theta \mathbf{e}_\theta,$$

$$\rho(\theta) = D_{II}, |_{\rho=r} = \sigma_0 + 2\varepsilon_0 E_a \cos \theta.$$

(a) The difference between this case and the case of the cylindrical conductor lies in the fact that $\sigma(\theta)$ can be positive or negative for a conductor, while in this case $\sigma(\theta) \leq 0$. However, when $|E_a| < \left| \frac{\sigma_0}{2\varepsilon_0} \right|$, the two problems have the same solution.

(b) For the case of a conducting cylinder the electrostatic field must satisfy the following:

(1) Inside the conductor $E_1 = 0$ and φ_1 is a constant.

(2) Outside the conductor

$$\nabla^2 \varphi_{II} = 0,$$

$$\left(\frac{\partial \varphi_{II}}{\partial \theta} \right)_{\rho=r} = 0, \quad \varphi_{II}|_{\rho \rightarrow \infty} \rightarrow -E_a \rho \cos \theta.$$

The solution for φ_{II} is the same as before. For the solution of the conductor to fit the case of an insulating cylinder, the necessary condition is $|E_a| \leq \left| \frac{\sigma_0}{2\varepsilon_0} \right|$, which ensures that the surface charge density on the cylinder is negative everywhere.

JUDGE RATIONALE

- **(a)** Correctly gives the conducting-cylinder surface charge density with θ measured from the applied-field direction and states the applicability condition that the electron-coated insulating cylinder must not require positive surface charge, equivalent to $|E_a| \leq |\sigma_0|/(2\epsilon_0)$ under the signed convention.

14. mechanics/1_27

PROBLEM STATEMENT

Three point particles with masses m_1, m_2 and m_3 interact with each other through the gravitational force.

- Write down the equations of motion.
- The system can rotate in its plane with constant and equal distances between all pairs of masses. Determine the angular frequency of the rotation when the masses are separated by a distance d .
- For $m_1 \gg m_3$ and $m_2 \gg m_3$, determine the stability condition for motion of the mass m_3 about the stationary position. Consider only motion in the orbital plane.

EXTRACTED ANSWER

- $\ddot{\mathbf{r}}_i = \sum_{j \neq i} Gm_j \frac{\mathbf{r}_j - \mathbf{r}_i}{|\mathbf{r}_j - \mathbf{r}_i|^3}, \quad i = 1, 2, 3.$
- $\omega = \sqrt{\frac{G(m_1 + m_2 + m_3)}{d^3}}.$
- For $m_3 \ll m_1, m_2$, the triangular stationary point is stable in the orbital plane if $\frac{m_1 m_2}{(m_1 + m_2)^2} < \frac{1}{27}$,
equivalently, if $\mu(1 - \mu) < \frac{1}{27}$, $\mu = \frac{m_2}{m_1 + m_2}.$

FINAL SOLUTION

Take the center of mass C of the system as the origin of coordinates and let the position vectors of m_1, m_2, m_3 be $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$ respectively as shown in Fig. 1.16. Denote

$$\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j \quad (i, j = 1, 2, 3).$$

- The motion of the i th particle is given by

$$m_i \ddot{\mathbf{r}}_i = - \sum_{j \neq i}^3 \frac{Gm_i m_j}{r_{ij}^3} \mathbf{r}_{ij},$$

or

$$\ddot{\mathbf{r}}_i = - \sum_{j \neq i}^3 \frac{Gm_j}{r_{ij}^3} \mathbf{r}_{ij} \quad (i = 1, 2, 3). \quad (1)$$

Note that the minus sign is to indicate that the forces are attractive.

(b) With the given condition $r_{ij} = d$, Eq. (1) is rewritten as

$$\begin{aligned}
 \ddot{\mathbf{r}}_i &= \frac{G}{d^3} \sum_{j \neq i}^3 m_j (\mathbf{r}_j - \mathbf{r}_i) \\
 &= \frac{G}{d^3} \left[- \sum_{j \neq i}^3 m_j \mathbf{r}_i + \sum_{j \neq i}^3 m_j \mathbf{r}_j \right] \\
 &= \frac{G}{d^3} \left[- \sum_{j \neq i}^3 m_j \mathbf{r}_i - m_i \mathbf{r}_i + \sum_{j=1}^3 m_j \mathbf{r}_j \right] \\
 &= \frac{G}{d^3} \left[- \mathbf{r}_i \sum_{j=1}^3 m_j + \sum_{j=1}^3 m_j \mathbf{r}_j \right] \\
 &= - \frac{GM}{d^3} \mathbf{r}_i,
 \end{aligned}$$

where $M = m_1 + m_2 + m_3$. Note that the choice of the center of mass as origin makes $\sum m_j \mathbf{r}_j$ vanish. Thus the force on each particle points towards the center of mass of the system and is a harmonic force. With d constant, the system rotates about C with angular frequency

$$\omega = \sqrt{\frac{GM}{d^3}}.$$

(c) For $m_3 \ll m_1$ and $m_3 \ll m_2$, the equation of motion of either of the masses m_1 and m_2 can be written as

$$\begin{aligned}
 \ddot{\mathbf{r}}_i &= - \frac{G(m_1 + m_2)}{d^3} \mathbf{r}_i \\
 &= - \frac{G(m_1 + m_2)}{r_{12}^3} \mathbf{r}_i, \quad i = 1, 2.
 \end{aligned}$$

With the distance between m_1 and m_2 constant, the system rotates about its center of mass with a constant angular frequency

$$\omega = \sqrt{\frac{G(m_1 + m_2)}{r_{12}^3}}.$$

Use a rotating coordinate frame with origin at the center of mass of the system and angular frequency of rotation ω and let the quantities $\ddot{\mathbf{r}}, \dot{\mathbf{r}}$ refer to this rotating frame. Considering the motion of particle m_3 in the laboratory frame, we have

$$m_3(\ddot{\mathbf{r}}_3 - \omega^2 \mathbf{r}_3) = -\frac{Gm_3m_1}{r_{31}^3} \mathbf{r}_{31} - \frac{Gm_3m_2}{r_{32}^3} \mathbf{r}_{32} - 2m_3\omega \times \dot{\mathbf{r}}_3,$$

or

$$\ddot{\mathbf{r}}_3 = -\frac{Gm_1}{r_{31}^3} \mathbf{r}_{31} - \frac{Gm_2}{r_{32}^3} \mathbf{r}_{32} + \omega^2 \mathbf{r}_3 - 2\omega \times \dot{\mathbf{r}}_3.$$

If m_3 is stationary, $\dot{\mathbf{r}}_3 = \ddot{\mathbf{r}}_3 = 0$ and the above becomes

$$\frac{Gm_1}{r_{31}^3}(\mathbf{r}_1 - \mathbf{r}_3) + \frac{Gm_2}{r_{32}^3}(\mathbf{r}_2 - \mathbf{r}_3) + \omega^2 \mathbf{r}_3 = 0.$$

With $m_1, m_2 \gg m_3$, $\sum m_j \mathbf{r}_j = 0$ gives $m_1 \mathbf{r}_1 \approx -m_2 \mathbf{r}_2$ and the above becomes

$$-G \left(\frac{m_1}{r_{31}^3} + \frac{m_2}{r_{32}^3} \right) \mathbf{r}_3 + G \left(\frac{m_1}{r_{31}^3} - \frac{m_1}{r_{32}^3} \right) \mathbf{r}_1 + \omega^2 \mathbf{r}_3 = 0.$$

This relation shows that $\mathbf{r}_3$ is parallel to $\mathbf{r}_1$ and thus the stationary position of m_3 lies on the line joining m_1 and m_2 . At this position, the attractions of m_1 and m_2 are balanced.

Consider now a small displacement being applied to m_3 at this stationary position. If the displacement is along the line joining m_1 and m_2 , say toward m_1 , the attraction by m_1 is enhanced and that by m_2 is reduced. Then m_3 will continue to move toward m_1 and the equilibrium is unstable. On the other hand, if the displacement is normal to the line joining m_1 and m_2 , both the attractions by m_1 and m_2 will have a component toward the stationary position and will restore m_3 to this position. Thus the equilibrium is stable. Therefore the equilibrium is stable against a transverse perturbation but unstable against a longitudinal one.

JUDGE RATIONALE

- **(a)** Correctly gives the three-body Newtonian equations of motion in equivalent vector form.
- **(b)** Correctly gives the angular frequency for equal pairwise separation d as $\sqrt{G(m_1+m_2+m_3)/d^3}$.
- **(c)** The reference solution describes the collinear stationary position for m_3 and states it is stable only for transverse perturbations and unstable for longitudinal perturbations. The candidate instead gives the triangular Lagrange-point stability criterion, which is not the referenced answer.

15. statistics/2-68

PROBLEM STATEMENT

Consider a photon gas enclosed in a volume V and in equilibrium at temperature T . The photon is a massless particle, so that $\epsilon = pc$.

- (a) What is the chemical potential of the gas?
- (b) Determine how the number of photons in the volume depends upon the temperature.
- (c) One may write the energy density in the form

$$\frac{E}{V} = \int_0^\infty \rho(\omega) d\omega.$$

Determine the form of $\rho(\omega)$, the spectral density of the energy.

- (d) What is the temperature dependence of the energy $\bar{E}$?

EXTRACTED ANSWER

$$(a) \mu = 0; \quad (b) N = \frac{2\zeta(3)}{\pi^2} V \left(\frac{k_B T}{\hbar c} \right)^3 \propto VT^3; \quad (c) \rho(\omega) = \frac{\hbar \omega^3}{\pi^2 c^3 (e^{\hbar \omega / k_B T} - 1)}; \quad (d) \bar{E} = \frac{\pi^2 k_B^4}{15 \hbar^3 c^3} VT^4 \propto VT^4.$$

FINAL SOLUTION

(a) The chemical potential of the photon gas is zero. Since the number of photons is not conserved at a given temperature and volume, the average photon number is determined by the expression $\left(\frac{\partial F}{\partial N} \right)_{T,V} = 0$, then

$$\mu = \left(\frac{\partial F}{\partial N} \right)_{T,V} = 0.$$

(b) The density of states is $8\pi V p^2 dp / \hbar^3$, or $V \omega^2 d\omega / \pi c^3$. Then the number of photons is

$$\bar{N} = \int \frac{V}{\pi^2 c^3} \omega^2 \frac{1}{e^{\hbar \omega / kT} - 1} d\omega = \frac{V}{\pi^2 c^3} \left(\frac{kT}{\hbar} \right)^3 \int_0^\infty \frac{\alpha^2 d\alpha}{e^\alpha - 1} \propto T^3.$$

(c), (d)

$$\frac{\bar{E}}{V} = \int \frac{\omega^2}{\pi^2 c^3} \frac{\hbar \omega}{e^{\hbar \omega / kT} - 1} d\omega = \frac{(kT)^4}{\pi^2 c^3 \hbar^3} \int \frac{\xi^3 d\xi}{e^\xi - 1}.$$

Hence

$$\rho(\omega) = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\hbar\omega/kT} - 1},$$

and $\bar{E} \propto T^4$.

JUDGE RATIONALE

- **(a)** Correct: the photon gas chemical potential is zero.
- **(b)** Correct: gives the photon number proportional to $V T^3$ with the standard coefficient equivalent to the reference integral.
- **(c)** Correct: the spectral energy density matches the Planck form in angular frequency used by the reference solution.
- **(d)** Correct: gives total energy proportional to $V T^4$ with the standard blackbody coefficient, consistent with the reference.

16. atomic/4-24

PROBLEM STATEMENT

State whether the following processes are possible or impossible:

- (a) A single photon strikes a stationary electron and gives up all its energy to the electron.
- (b) A single photon in empty space is transformed into an electron and a positron.
- (c) A fast positron and a stationary electron annihilate, producing only one photon.

EXTRACTED ANSWER

(a) Impossible; (b) Impossible; (c) Impossible.

FINAL SOLUTION

All the three reactions cannot take place because in each case energy and momentum cannot be both conserved.

(a) For the process

$$\gamma + e \rightarrow e',$$

conservation of the invariant mass squared,

$$S = (E_\gamma + m_e)^2 - p_\gamma^2 = 2m_e E_\gamma + m_e^2 = E_{e'}^2 - p_{e'}^2 = m_e^2,$$

leads to $m_e E_\gamma = 0$, which contradicts the fact that neither E_γ nor m_e is zero.

(b) In the process $\gamma \rightarrow e^+ + e^-$, let the energies and momenta of the produced e^+ and e^- be $E_1, E_2, \mathbf{p}_1, \mathbf{p}_2$ respectively. The invariant mass squared of the initial state is

$$S(\gamma) = E_\gamma^2 - p_\gamma^2 = 0,$$

while for the final state it is

$$S(e^+e^-) = (E_1 + E_2)^2 - (\mathbf{p}_1 + \mathbf{p}_2)^2 = 2m_e^2 + 2(E_1 E_2 - p_1 p_2 \cos \theta) \geq 2m_e^2,$$

where θ is the angle between $\mathbf{p}_1$ and $\mathbf{p}_2$. As $S(\gamma) \neq S(e^+e^-)$, its invariance is violated and the reaction cannot take place.

(c) The reaction is the inverse of that in (b). It similarly cannot take place.

JUDGE RATIONALE

- **(a)** Correctly states the process is impossible, matching the reference result that energy and momentum cannot both be conserved.
- **(b)** Correctly states that a single photon in empty space cannot transform into an electron-positron pair.
- **(c)** Correctly states that annihilation into only one photon is impossible, matching the reference.

17. optics/1-23

PROBLEM STATEMENT

As shown in Fig. 1.29, the image which would be cast by the converging lens alone has a distance of 0.5 cm between top and bottom.

Calculate, of the final image:

(a) the position.

(b) size.

EXTRACTED ANSWER

(a) Underdetermined without Fig. 1.29; generally $v = \frac{fu}{u-f}$ from the second lens. (b) Underdetermined; generally $h_f = 0.5 \left| \frac{f}{u-f} \right|$ cm.

FINAL SOLUTION

An object infinite distance away finds its image on the back focal plane of the converging lens, i.e., at $20 - 15 = 5$ cm behind the diverging lens. The Gaussian lens formula

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

applied to the diverging lens then gives for $u = -5$ cm and $f = -10$ cm, $v = 10$ cm. The lateral magnification is

$$m = \frac{h}{h'} = \frac{10}{5} = 2.$$

Hence the size of the final image is

$$0.5 \times 2 = 1 \text{ cm}.$$

Fig. 1.30 gives the ray diagram:

JUDGE RATIONALE

- **(a)** Reference gives a definite final image position of $v = 10$ cm for the diverging lens; candidate instead says underdetermined and gives only a general formula.
- **(b)** Reference gives final image size 1 cm; candidate instead says underdetermined and gives only a general expression.

18. quantum/2-2001

PROBLEM STATEMENT

An electron is confined in a three-dimensional infinite potential well. The sides parallel to the x -, y -, and z -axes are of length L each.

- Write the appropriate Schrödinger equation.
- Write the time-independent wave function corresponding to the state of the lowest possible energy.
- Give an expression for the number of states, N , having energy less than some given E . Assume $N \gg 1$.

EXTRACTED ANSWER

$$\begin{aligned}
 \text{(a)} \quad & -\frac{\hbar^2}{2m_e} \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right) = E\psi, \quad 0 < x, y, z < L, \\
 & \psi = 0 \text{ on } x, y, z = 0, L. \\
 \text{(b)} \quad & \psi_{111}(x, y, z) = \left(\frac{2}{L} \right)^{3/2} \sin \frac{\pi x}{L} \sin \frac{\pi y}{L} \sin \frac{\pi z}{L}, \\
 & E_{111} = \frac{3\pi^2 \hbar^2}{2m_e L^2}. \\
 \text{(c)} \quad & E_{n_x n_y n_z} = \frac{\pi^2 \hbar^2}{2m_e L^2} (n_x^2 + n_y^2 + n_z^2), \\
 & N(E) \simeq \frac{1}{8} \frac{4\pi}{3} \left(\frac{2m_e L^2 E}{\pi^2 \hbar^2} \right)^{3/2} = \frac{L^3}{6\pi^2} \left(\frac{2m_e E}{\hbar^2} \right)^{3/2}. \text{ Including electron spin degeneracy, multiply by 2.}
 \end{aligned}$$

FINAL SOLUTION

- (a) The Schrödinger equation is

$$i\hbar \partial \psi(r, t) / \partial t = -(\hbar^2 / 2m) \nabla^2 \psi(r, t), \quad 0 \leq x, y, z \leq L,$$

$$\psi = 0, \quad \text{otherwise.}$$

- (b) By separation of variables, we can take that the wave function to be the product of three wave functions each of a one-dimensional infinite well potential. The wave function of the lowest energy level is

$$\psi_{111}(x, y, z) = \psi_1(x)\psi_1(y)\psi_1(z),$$

where

$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi}{L}x\right), \quad \text{etc.}$$

Thus

$$\psi_{111}(x, y, z) = \left(\frac{2}{L}\right)^{3/2} \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi y}{L}\right) \sin\left(\frac{\pi z}{L}\right).$$

The corresponding energy is $E_{111} = 3\hbar^2\pi^2/2mL^2$.

(c) For a set of quantum numbers n_x, n_y, n_z for the three dimensions, the energy is

$$E = \frac{\hbar^2\pi^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2).$$

Hence the number N of states whose energy is less than or equal to E is equal to the number of sets of three positive integers n_x, n_y, n_z satisfying the inequality

$$n_x^2 + n_y^2 + n_z^2 \leq \frac{2mL^2E}{\hbar^2\pi^2}.$$

Consider a Cartesian coordinate system of axes n_x, n_y, n_z . The number N required is numerically equal to the volume in the first quadrant of a sphere of radius $(2mL^2E/\hbar^2\pi^2)^{1/2}$, provided $N \geq 1$. Thus

$$N = \frac{1}{8} \cdot \frac{4\pi}{3} \cdot \left(\frac{2mL^2}{\hbar^2\pi^2}E\right)^{3/2} = \frac{4\pi}{3} \left(\frac{mL^2}{2\hbar^2\pi^2}E\right)^{3/2}.$$

JUDGE RATIONALE

- **(a)** The candidate gives the correct time-independent Schrödinger equation for a free electron inside the 3D infinite well with appropriate boundary conditions; this is an acceptable equivalent stationary-state form of the reference equation.
- **(b)** The candidate gives the correctly normalized ground-state wave function with quantum numbers (1,1,1), and the stated energy is also correct.

- **(c)** The candidate gives the correct energy spectrum and the large-N first-octant sphere counting formula, equivalently simplified to the standard density-of-states expression; noting optional spin degeneracy does not materially change the requested spatial-state count.

19. Quantum Mechanics/27-5

PROBLEM STATEMENT

Positronium is an atom made up of an electron and a positron (electron anti-particle).

(a) As a "zeroth" order approximation, neglect everything (e.g., fine structure, hyperfine structure, etc.) except the Coulombic interaction between the electron and positron. Relate the "zeroth" order positronium energies to those of hydrogen.

(b) Taking spin into account, each of the levels of part (a) are 4-degenerate (since the electron and positron are both spin-1/2 particles). Taking spin-spin interactions into account, $H = H_0 - A\mu_p \cdot \mu_e$, where μ is the magnetic moment, p stands for positron and e for electron. Place the positronium in a uniform and constant external magnetic field,

B. What are the resulting energies of a level?

EXTRACTED ANSWER

$$(a) E_{n,Ps}^{(0)} = \frac{\mu_{Ps}}{\mu_H} E_{n,H}^{(0)}, \quad \mu_{Ps} = \frac{m_e}{2}, \quad \mu_H = \frac{m_e M_p}{m_e + M_p}.$$

$$\text{Thus } E_{n,Ps}^{(0)} \simeq \frac{1}{2} E_{n,H}^{(0)} = -\frac{6.80 \text{ eV}}{n^2}.$$

$$(b) \text{ For a fixed zeroth-order level } E_0 = E_{n,Ps}^{(0)}, \text{ with } \mu = g\mu_B/2,$$

$$E_{1,+1} = E_{1,-1} = E_0 + A\mu^2,$$

$$E_{\pm} = E_0 - A\mu^2 \pm \sqrt{(2A\mu^2)^2 + (2\mu B)^2}.$$

FINAL SOLUTION

(a) The Bohr Hydrogen energies are $E_n = -\frac{1}{2n^2}\alpha^2 m_e c^2$, where $\alpha = e^2/4\pi\epsilon_0\hbar c \approx 1/137$ is the fine structure constant. To correct for finite proton mass, one uses the reduced electron mass $\mu = \frac{m_e m_p}{m_e + m_p}$. In this case, $m_p = m_e$, so $\mu = m_e/2$. Thus, *Positronium energies are half of Bohr energies*.

(b) $\mu = \gamma S$, where γ is gyromagnetic ratio. $\gamma = -g\mu_B/\hbar$, where g is the g-factor. We have $g_e = -g_p = 2$ and so $\mu_e = -\frac{2\mu_B}{\hbar} S_e$, and $\mu_p = \frac{2\mu_B}{\hbar} S_p$. The Hamiltonian is

$$\begin{aligned} H &= H_0 - A\mu_p \cdot \mu_e - (\mu_e + \mu_p) \cdot \mathbf{B} \\ &= H_0 + \left(\frac{2\mu_B}{\hbar}\right)^2 A \mathbf{S}_p \cdot \mathbf{S}_e + \left(\frac{2\mu_B}{\hbar}\right) B (S_e^z - S_p^z) \end{aligned}$$

(27.5.1)

Let $|1\rangle = |\uparrow\uparrow\rangle$, $|2\rangle = |\downarrow\downarrow\rangle$, $|3\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$ and $|4\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$, where the first spin is the electron and the second the positron. In the $|Sm_S\rangle$ basis, $|1\rangle = |11\rangle$, $|2\rangle = |1, -1\rangle$, $|3\rangle = |10\rangle$, $|4\rangle = |00\rangle$.

$\mathbf{S}_p \cdot \mathbf{S}_e = \frac{1}{2} (S^2 - S_e^2 - S_p^2) = \frac{\hbar^2}{4} \left(\frac{2S^2}{\hbar^2} - 3 \right)$ which is $\frac{\hbar^2}{4}$ for $S = 1$ and $-\frac{3}{4}\hbar^2$ for $S = 0$. Thus, $\left(\frac{2\mu_B}{\hbar}\right)^2 S_p \cdot S_e$ has the value μ_B^2 on $S = 1$ states and $-3\mu_B^2$ on $S = 0$ states.

Write the magnetic field term as $B\mu_B (\sigma_e^z - \sigma_p^z)$. It is clear that this operator vanishes on $|1\rangle$ and $|2\rangle$. However, $\sigma_e^z|3\rangle = |4\rangle$ and $\sigma_p^z|3\rangle = -|4\rangle$, etc. Hence, in this basis, the Hamiltonian reads

$$H = \begin{pmatrix} H_0 + A\mu_B^2 & 0 & 0 & 0 \\ 0 & H_0 + A\mu_B^2 & 0 & 0 \\ 0 & 0 & H_0 + A\mu_B^2 & 2B\mu_B \\ 0 & 0 & 2B\mu_B & H_0 - 3A\mu_B^2 \end{pmatrix}$$

(27.5.2)

Forget H_0 since that does not affect diagonalization. We must diagonalize $\begin{pmatrix} 1 & 2\beta \\ 2\beta & -3 \end{pmatrix}$, where $\beta = B/A\mu_B$. The secular equation reads $(\lambda - 1)(\lambda + 3) - 4\beta^2 = \lambda^2 + 2\lambda - (3 + 4\beta^2) = 0$ with solutions $\lambda_{\pm} = -1 \pm \sqrt{1 + 3 + 4\beta^2} = -1 \pm 2\sqrt{1 + \beta^2}$. Thus,

1. $|1\rangle$ and $|2\rangle$ remain unchanged with energy $H_0 + A\mu_B^2$,
2. $|3\rangle$ transforms with energy $H_0 + A\mu_B^2 + 2A\mu_B^2 \left[\sqrt{1 + \beta^2} - 1 \right]$,
3. $|4\rangle$ transforms with energy $H_0 - 3A\mu_B^2 - 2A\mu_B^2 \left[\sqrt{1 + \beta^2} - 1 \right]$.

(27.5.3)

JUDGE RATIONALE

- **(a)** Correctly states that positronium energies scale with the reduced mass and are approximately half the hydrogen Bohr energies, giving the correct $-6.80 \text{ eV}/n^2$ form.
- **(b)** Gives the two unchanged triplet energies and the two mixed $m=0$ energies in a form equivalent to diagonalizing the reference 2×2 Hamiltonian, with $\mu = \mu_B$ for $g = 2$.

20. electro/1_45

PROBLEM STATEMENT

Two conductors are embedded in a material of conductivity $10^{-4} \Omega/m$ and dielectric constant $\varepsilon = 80\varepsilon_0$. The resistance between the two conductors is measured to be $10^5 \Omega$.

- (a) Derive an equation for the capacitance between the two conductors.
 (b) calculate its value.

EXTRACTED ANSWER

$$(a) C = \frac{\varepsilon}{\sigma R} \quad (b) C = 7.08 \times 10^{-11} \text{ F} \approx 70.8 \text{ pF}$$

FINAL SOLUTION

Solution:

Suppose that the two conductors carry free charges Q and $-Q$. Consider a closed surface enclosing the conductor with the charge Q (but not the other conductor). We have, using Ohm's and Gauss' laws,

$$I = \oint \mathbf{j} \cdot d\mathbf{S} = \oint \sigma \mathbf{E} \cdot d\mathbf{S} = \sigma \oint \mathbf{E} \cdot d\mathbf{S} = \sigma \frac{Q}{\varepsilon}.$$

If the potential difference between the two conductors is V , we have $V = IR = \frac{\sigma Q}{\varepsilon} R$, whence

$$C = \frac{Q}{V} = \frac{\varepsilon}{\sigma R}.$$

Numerically the capacitance between the conductors is

$$C = \frac{80 \times 8.85 \times 10^{-12}}{10^{-4} \times 10^5} = 7.08 \times 10^{-11} \text{ F}.$$

JUDGE RATIONALE

- **(a)** Gives the correct relation $C = \varepsilon/(\sigma R)$ for the capacitance between the conductors.
- **(b)** Computes the capacitance as $7.08 \times 10^{-11} \text{ F}$, equivalent to the reference value; 70.8 pF is a correct conversion.